Towards an understanding of avian malaria parasite evolution

The draft genome of *Haemoproteus tartakovskyi and some bioinformatic stuff*

Björn Canbäck, Staffan Bensch
Department of Biology
Lund University

Outline

1. The genome of *Haemoproteus tartakovskyi*.
2. New algorithms that need to be developed.

Don’t become a malaria researcher

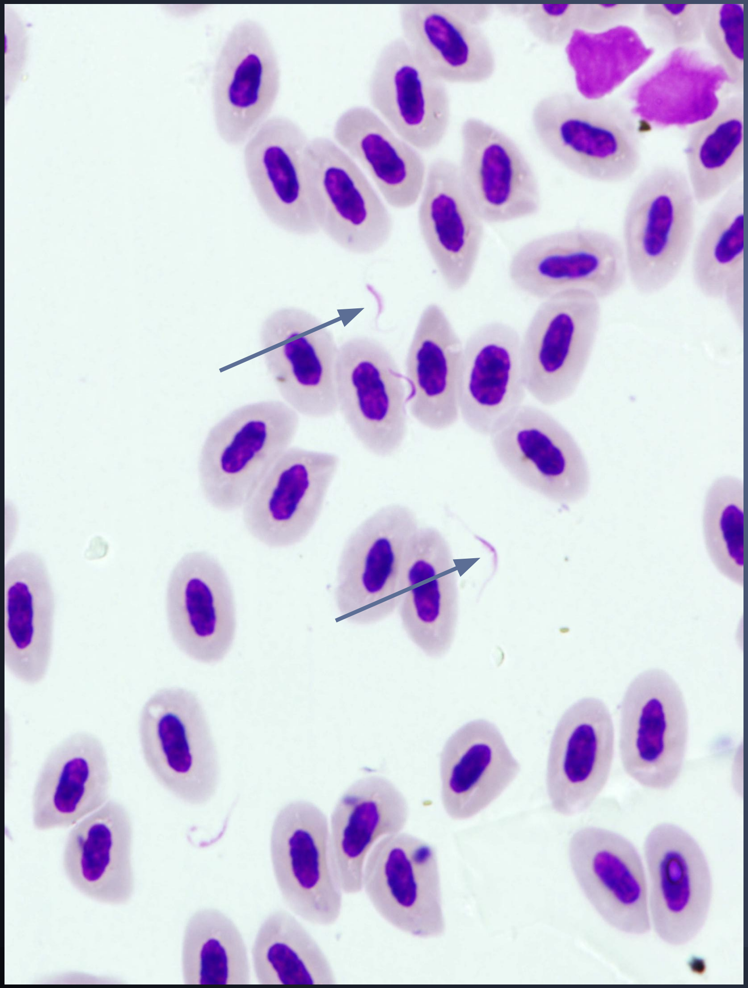
sporozoite	vector salivary glands
schizonts	host, blood -> tissues
merozoites	tissues -> erythrocytes
gametozytes	erythrocytes -> Vector
- macrogametocytes	female
- microgametocytes	male
oocysts	vector midgut, sexual reproduction

Differences between *Plasmodium* and *Haemoproteus*

- *Plasmodium* replicate asexually in blood cells (and other cells). In *Haemoproteus* only the gametocytes use the blood cells.
- Avian parasites are less host specific than mammalian parasites
- Different vectors (mosquitoes vs. midges)
- Bird (and reptile) blood cells have nuclei.

"Genomic data in itself does little more than clogging up computers”

Francois Balloux in a job advertisement



One problem:

Bird genome size: 1100 Mb

Parasite genome size: 25 Mb

Ratio: 2%

Two solutions:

1) Enrich microgamets and centrifuge

2) Use differential GC content

Bird: 41 %

Parasite: 19-42%

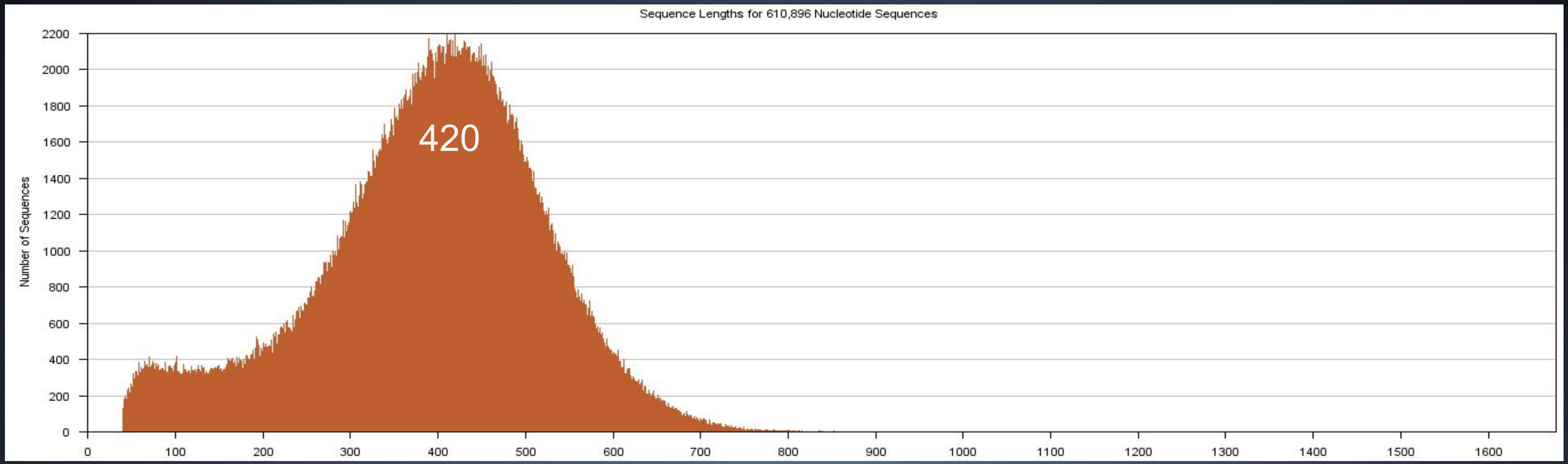
Important questions

1. What are the phylogenetic relations between *Haemoprotues* and *Plasmodium*?
2. Is the asexual replication in blood cells a gain in *Plasmodium* or a loss in *Haemoproteus*?
3. What is the broader host range for avian malaria parasites due to?

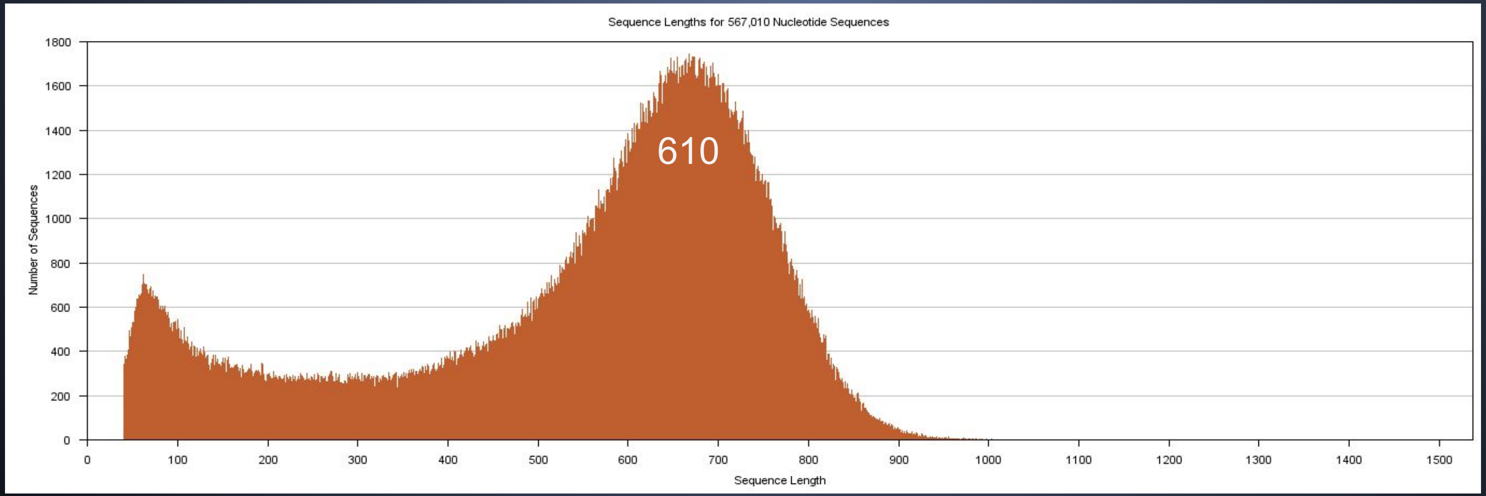
Haemoproteus tartakovskyi

Sequenced using 454-pyrosequencing

Shotgun 1.2 million reads:

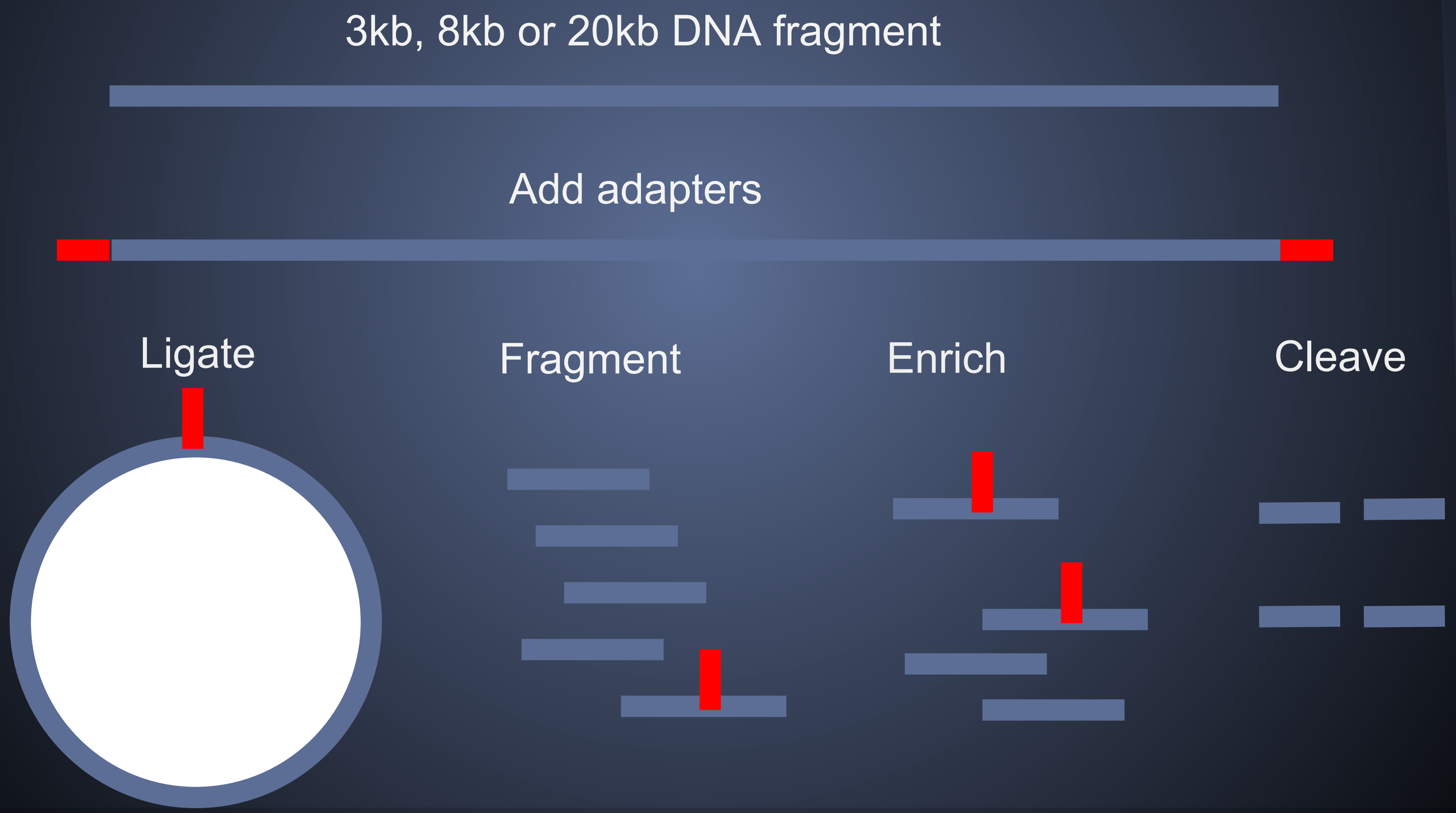


Shotgun 0.6 million reads:

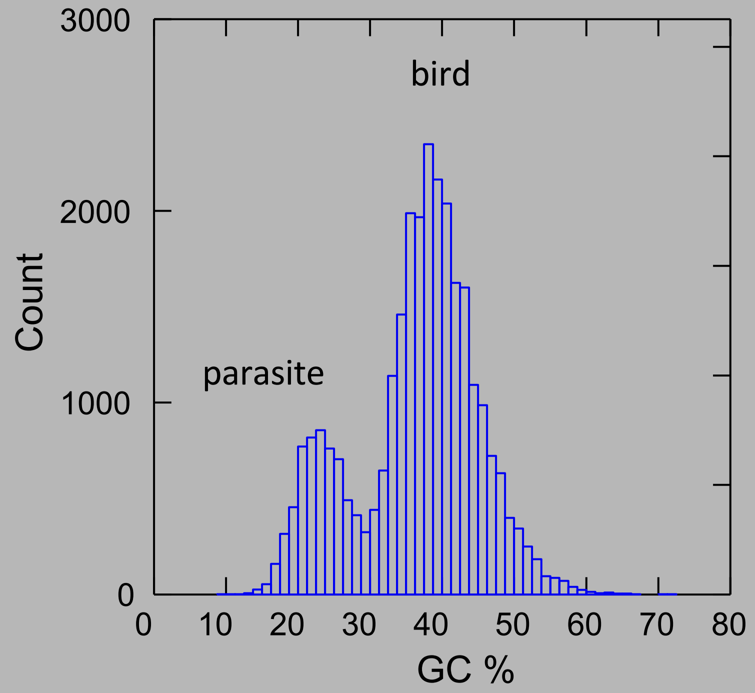


+ Paired end 1.2 million reads

Paired end (454 example)



Roche runAssembly software of all reads => 28,500 contigs



Read removal

BLAT of contigs to zebra finch and *Plasmodium* removed 1.1 million reads.
Compare with previous slide, is this correct?
The remaining 1.9 million reads were assembled again.

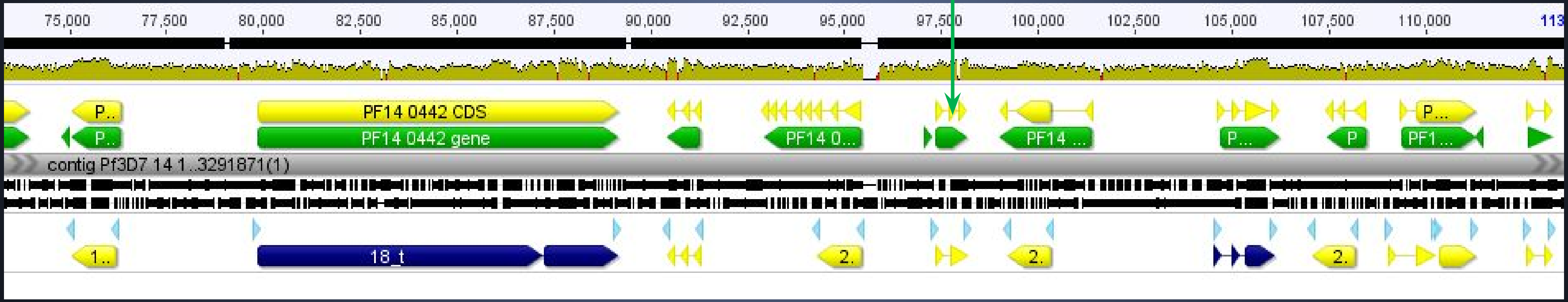
- Number of scaffolds: 2,233
- Total nucleotides without N's: 21,824,431
- Longest scaffold: 116,706
- N50: 16,618
- GC: 26.00%

The assembly still contains some (<5% ?) zebra finch sequences and missing some tartakovskyi

The de novo ”Gene Prediction” worked amazingly well ...

Parts of Scaffold00001 – *P. falciparum* Chr14

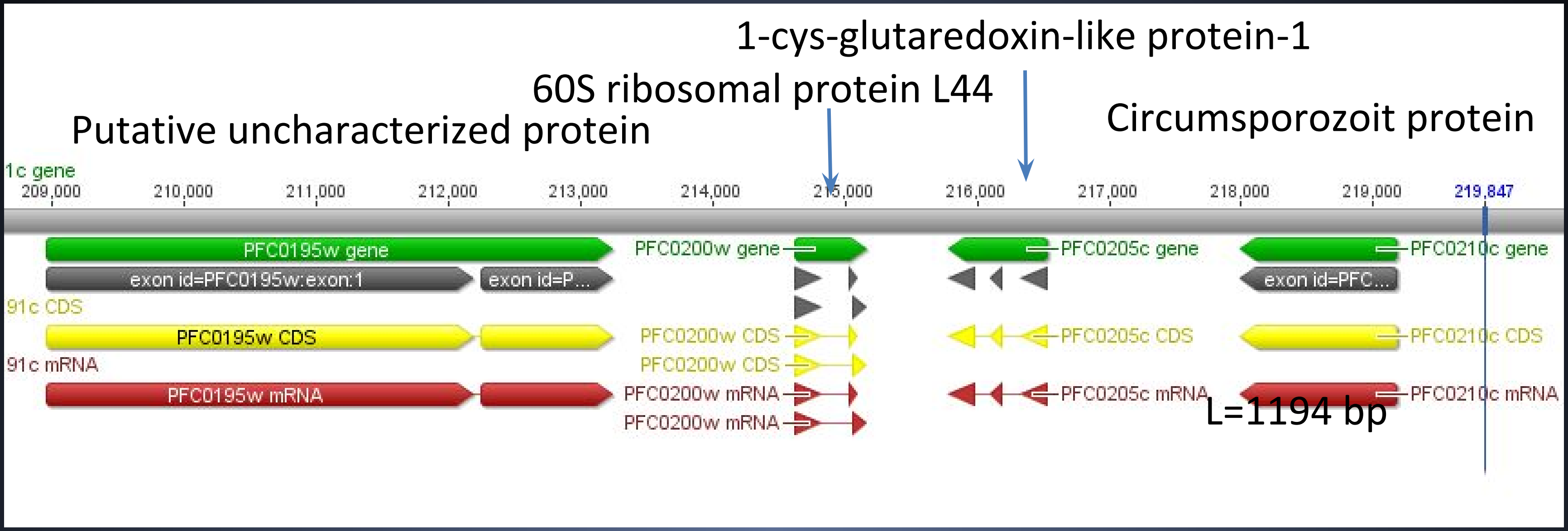
Heme detoxification protein



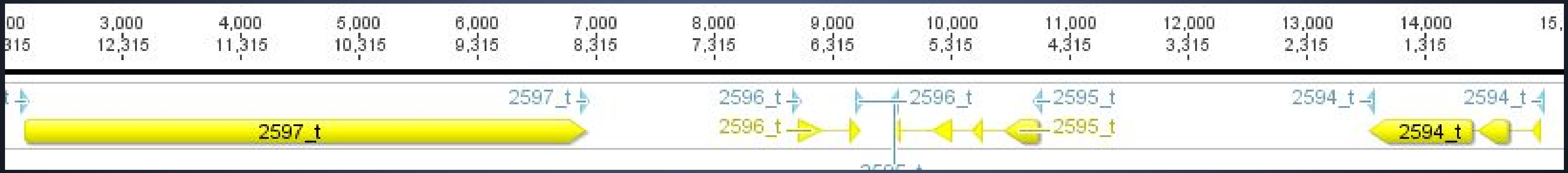
Apicomplexans

Species	**Species Ab.**	**Mb**	**Protein #**	**Chr #**	**Scaffolds #**
Plasmodium falciparum	*Pf*	~23	5538	14	14
P. knowlesi	*Pk*	~23	5226	14	81
P. vivax	*Pv*	~25	5393	14	2778
P. cynomolgi	*Pm*	~25	5716	14	1663
P. berghei	*Pb*	~23	5012	14	15
P. chabaudi	*Pc*	~23	5166	14	15
P. yoellii (YM)	*Py*	~23	5710	14	15
Haemoproteus tartakovskyi	*Ht*	~23	5519	?	2233
Babesia bovis	*Bb*	~9	3671	4	12
B. microti	*Bm*	~6	3494	3	3493
Theileria annulata	*Ta*	~8.5	3796	4	9
T. parva	*Tp*	~8.5	4023	4	10
Toxoplasma gondii (ME49)	*Tg*	~63	8320	14	2006
Neospora caninum	*Nc*	~62	7122	14	66
Sarcocystis neurona	*Sn*	~120	13332	?	172
Eimeria tenella	*Et*		8786	14	4681
Cryptosporidium muris	*Cm*	~9.2	3937	8	45
C. parvum	*Cp*	~9.1	3805	8	9

P. falciparum Chr 3



H. tartakovskyi scaffold 417

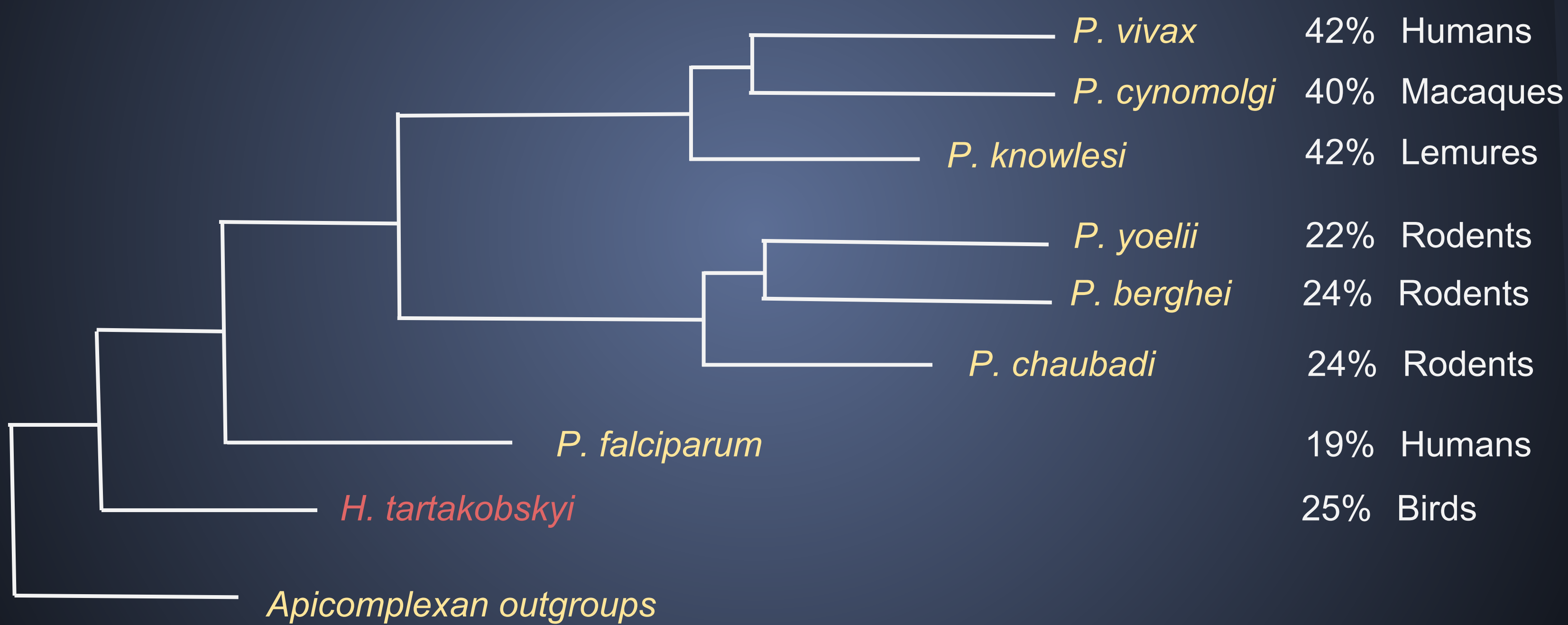


E-18 E-35 E-49

Phylogeny

- Difficult to do phylogenetic trees when base composition is biased.
- Typically, concatenate aligned genes and remove third codon position. Use logdet or similar method for substitution.
- Here, we first found out orthologs by using CEGMA and ORTHO-MCL.
- We used individual protein alignments and summed up the support for each node.
- We also concatenated the protein alignments.

A tree of 703 protein families



Thank you!

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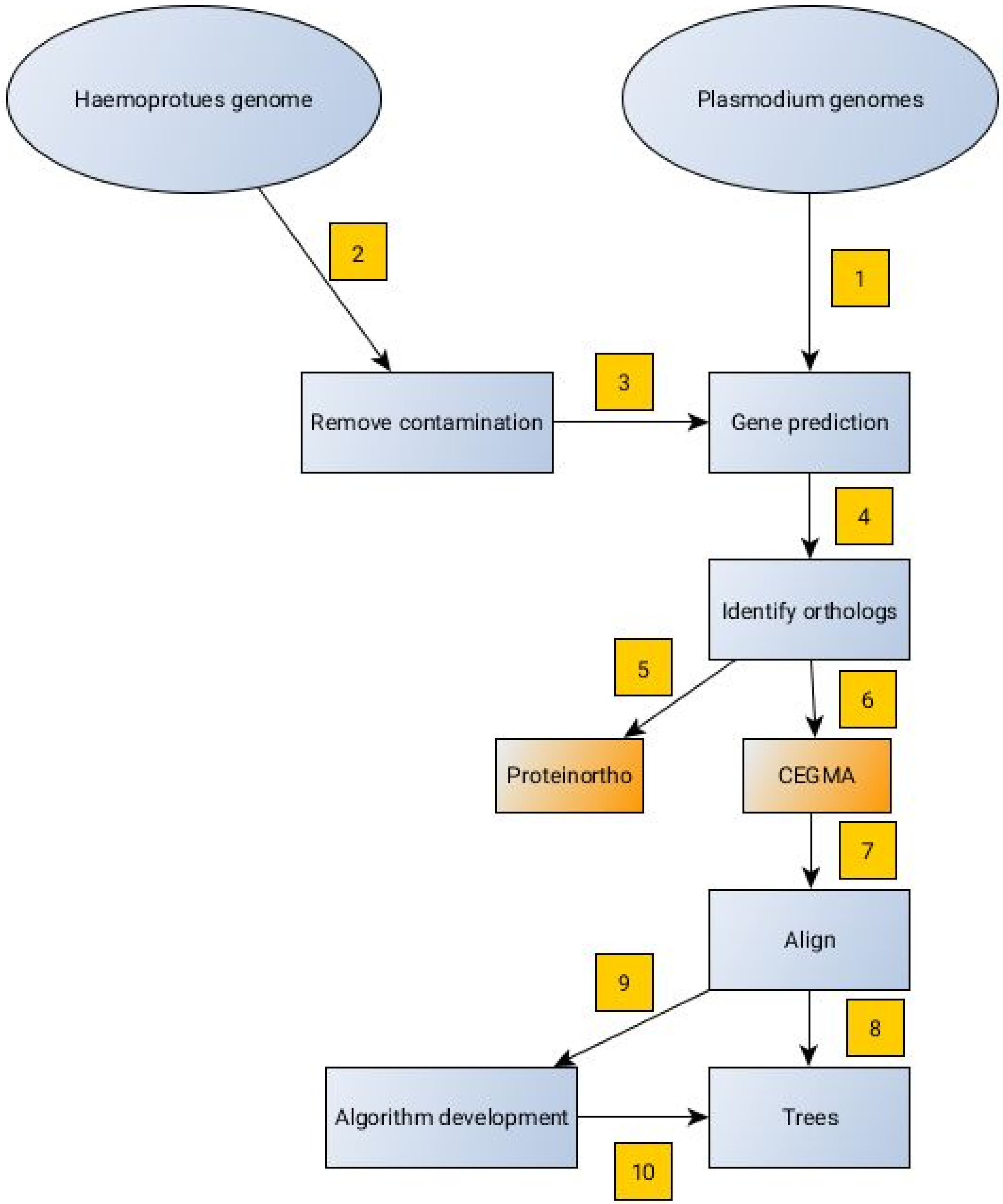
Haemoprotues genome
Plasmodium genomes
1
2
Remove contamination
3
Gene prediction
4
Identify orthologs
5
6
Proteinortho
CEGMA
7
Align
9
8
Algorithm development
Trees
10